

LESLIE KADENGE

AI Engineer

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SUMMARY

Information Technology graduate working through a public portfolio of retrieval and applied AI projects, each with reproducible code and a measured result. Every figure below can be reproduced from the linked repository by running one script.

The through-line is evaluation rather than model building: establishing a baseline before claiming a result, testing whether an apparent improvement survives a significance test, and reporting the cases where it does not.

PROJECTS

Semantic Search Python · sentence-transformers · information retrieval

github.com/les-k/semantic-search

- Compared six retrieval methods over 1,400 documents against 225 queries and 1,837 human relevance judgments.
- Hybrid retrieval — reciprocal rank fusion of BM25 and neural embeddings — reached recall@10 of 0.437 against 0.377 for BM25 alone, and 0.010 for random ranking.
- Applied paired significance testing to every claimed improvement: two of five did not survive, including one with a higher average on every metric ($p = 0.895$).
- Implemented BM25 directly from the formula, and identified three defects in the benchmark's relevance judgments that would each have corrupted the metrics silently.

Rafiki — voice assistant in Kiswahili Python · speech recognition · Chrome extension

github.com/les-k/Rafiki-Digital-assistant-AI

- A voice assistant whose entire command vocabulary is Kiswahili — the wake word, the verbs and the questions — rather than an English assistant with translation added in front.
- As of 2026 Alexa supports 9 languages, Siri around 21 and Google Assistant roughly 30; none support Swahili, a language with about 200 million speakers.
- Roughly 650 lines of Python for speech recognition, language detection and synthesis, paired with a Chrome extension that reaches what a desktop process cannot.

Credit Card Fraud Detection Python · scikit-learn · FastAPI · Docker

github.com/les-k/fraud-detection

- 492 frauds in 284,807 transactions — 0.17%. Established up front that a model flagging nothing scores 99.87% accuracy, making accuracy unusable as a metric.
- Set the decision threshold from an explicit cost model rather than the 0.50 default, giving 0.05 and recovering \$12,168 across a 48-hour holdout at 82% recall and 53% precision.
- Served as a containerised FastAPI endpoint with the trained model committed at 1 MB, so the API runs without the 150 MB dataset.

Recommendation Engine Python · scikit-learn · collaborative filtering

github.com/les-k/Recommendation-Engine

- User-based collaborative filtering over a sparse user-item matrix with cosine similarity.
- Documents the three defects the naive version carries: an unseen item treated as a zero rating, candidates ordered by id rather than by any score, and a row lookup that works only because the sample data happens to be sorted.

TECHNICAL SKILLS

LANGUAGES Python, SQL, JavaScript

MACHINE LEARNING scikit-learn, sentence-transformers, TensorFlow; classification, clustering, retrieval

RETRIEVAL BM25, TF-IDF, LSA, neural embeddings, reciprocal rank fusion; nDCG, MRR, recall@k

EVALUATION Baselines, significance testing, cost-sensitive thresholds, experiment design

DATA pandas, NumPy, SciPy, DuckDB

ENGINEERING FastAPI, Docker, Git, GitHub Actions, pytest

ACADEMIC PROJECT

MOODS — a sentiment analysis model built as a final-year project, classifying posts as positive, negative or neutral.

EDUCATION

BSc Information Technology — Zetech University, November 2023 Second-class honours, upper division

Coursework: Machine Learning, Data Science, Artificial Intelligence, Database Systems, Software Engineering, Distributed Systems, Programming Fundamentals (Python, Java, C, C++, JavaScript).

CERTIFICATION

ICDL Profile — Computer Pride.

References available on request.